



PROBLEM SESSION 7

Processor Design

คณะผู้จัดทำ

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รายงานนี้เป็นส่วนหนึ่งของรายวิชา

CPE223 สถาปัตยกรรมคอมพิวเตอร์ (Computer Architectures)

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มหาวิทยาลัยเทคโนโลยีพระจอมเกล้าธนบุรี

ภาคเรียนที่ 2 ปีการศึกษา 2568

Task 1

Instructions
ADD R_n , off(R_n) ADD R_n , #value ADD R_n , R_n MOV R_n , (R_n) MOV (index), R_n BZ (R_n) BZ +#relative B #Absolute

Execution Steps: ADD R_n , off(R_n)

Step	Action	Explanation
Fetch Stage		
1	PC_{out} , MAR_{in} , Read, Select4, Add, Z_{in}	Send PC to MAR, read memory, calculate $PC + 4$
2	Z_{out} , PC_{in} , WMFC	Update PC, wait for memory read to complete
3	MDR_{out} , IR_{in}	Transfer instruction from MDR to IR
Execution Stage		
4	$R_{n,out}$, Y_{in}	Move R_n to register Y
5	$IR(offset)_{out}$, SelectY, Add, Z_{in}	Add offset to R_n , store effective address in Z
6	Z_{out} , MAR_{in} , Read	Send effective address to MAR, read memory
7	$R_{n,out}$, Y_{in} , WMFC	Move R_n to Y, wait for memory read
8	MDR_{out} , SelectY, Add, Z_{in}	Add fetched data to R_n , store in Z
9	Z_{out} , $R_{n,in}$, End	Store final sum in R_n , end execution

Execution Steps: ADD $R_n, \#value$

Step	Action	Explanation
Fetch Stage		
1	$PC_{out}, MAR_{in}, Read, Select4, Add, Z_{in}$	Send PC to MAR, read memory, calculate $PC + 4$
2	$Z_{out}, PC_{in}, WMFC$	Update PC, wait for memory read to complete
3	MDR_{out}, IR_{in}	Transfer instruction from MDR to IR
Execution Stage		
4	$R_{n,out}, Y_{in}$	Move R_n to register Y
5	$IR(immediate)_{out}, SelectY, Add, Z_{in}$	Add immediate value to R_n , store in Z
6	$Z_{out}, R_{n,in}, End$	Store final sum in R_n , end execution

Execution Steps: ADD R_n, R_n

Step	Action	Explanation
Fetch Stage		
1	$PC_{out}, MAR_{in}, Read, Select4, Add, Z_{in}$	Send PC to MAR, read memory, calculate $PC + 4$
2	$Z_{out}, PC_{in}, WMFC$	Update PC, wait for memory read to complete
3	MDR_{out}, IR_{in}	Transfer instruction from MDR to IR
Execution Stage		
4	$R_{n,out}, Y_{in}$	Move R_n to register Y
5	$R_{n,out}, SelectY, Add, Z_{in}$	Add R_n to R_n (from Y), store in Z
6	$Z_{out}, R_{n,in}, End$	Store final sum in R_n , end execution

Execution Steps: MOV $R_n, (R_n)$

Step	Action	Explanation
Fetch Stage		
1	$PC_{out}, MAR_{in}, Read, Select4, Add, Z_{in}$	Send PC to MAR, read memory, calculate $PC + 4$
2	$Z_{out}, PC_{in}, WMFC$	Update PC, wait for memory read to complete
3	MDR_{out}, IR_{in}	Transfer instruction from MDR to IR
Execution Stage		
4	$R_{n,out}, MAR_{in}, Read$	Send R_n (address) to MAR, read memory
5	$WMFC$	Wait for memory read to complete
6	$MDR_{out}, R_{n,in}, End$	Transfer fetched data to R_n , end execution

Execution Steps: MOV (index), R_n

Step	Action	Explanation
Fetch Stage		
1	$PC_{out}, MAR_{in}, Read, Select4, Add, Z_{in}$	Send PC to MAR, read memory, calculate $PC + 4$
2	$Z_{out}, PC_{in}, WMFC$	Update PC, wait for memory read to complete
3	MDR_{out}, IR_{in}	Transfer instruction from MDR to IR
Execution Stage		
4	$IR(address)_{out}, MAR_{in}$	Send absolute address to MAR
5	$R_{n,out}, MDR_{in}$	Transfer R_n data to MDR
6	Write, End	Write MDR data to memory, end execution

Execution Steps: BZ (R_n)

Step	Action	Explanation
Fetch Stage		
1	$PC_{out}, MAR_{in}, Read, Select4, Add, Z_{in}$	Send PC to MAR, read memory, calculate $PC + 4$
2	$Z_{out}, PC_{in}, WMFC$	Update PC, wait for memory read to complete
3	MDR_{out}, IR_{in}	Transfer instruction from MDR to IR
Execution Stage		
4	If $Z = 1$ then $R_{n,out}, PC_{in}$	If Zero flag is set, load PC with R_n (target address)
5	End	End execution

Execution Steps: BZ + #relative

Step	Action	Explanation
Fetch Stage		
1	PC _{out} , MAR _{in} , Read, Select4, Add, Z _{in}	Send PC to MAR, read memory, calculate PC + 4
2	Z _{out} , PC _{in} , WMFC	Update PC, wait for memory read to complete
3	MDR _{out} , IR _{in}	Transfer instruction from MDR to IR
Execution Stage		
4	PC _{out} , Y _{in}	Move PC to register Y
5	IR(offset) _{out} , SelectY, Add, Z _{in}	Add PC and offset, store branch target in Z
6	If Z = 1 then Z _{out} , PC _{in}	If Zero flag is set, update PC with branch target
7	End	End execution

Execution Steps: B #Absolute

Step	Action	Explanation
Fetch Stage		
1	PC _{out} , MAR _{in} , Read, Select4, Add, Z _{in}	Send PC to MAR, read memory, calculate PC + 4
2	Z _{out} , PC _{in} , WMFC	Update PC, wait for memory read to complete
3	MDR _{out} , IR _{in}	Transfer instruction from MDR to IR
Execution Stage		
4	IR(address) _{out} , PC _{in}	Load absolute address directly into PC
5	End	End execution

- Jump to new address by replace **Program Counter (PC)** with address **#Absolute**

Task 2

Identify when each control signal is used (Only Add and Branch)

Instruction	Step	Micro-operation
Fetch cycle	T1	PC_{out} , MAR_{in} , Read, Select4, Add, Z_{in}
	T2	Z_{out} , PC_{in} , WMFC
	T3	MDR_{out} , IR_{in}
ADD $R_n, off(R_n)$	T4	$R_{n,out}$, Y_{in}
	T5	$IR(offset)_{out}$, SelectY, Add, Z_{in}
	T6	Z_{out} , MAR_{in} , Read
	T7	$R_{n,out}$, Y_{in} , WMFC
	T8	MDR_{out} , SelectY, Add, Z_{in}
	T9	Z_{out} , $R_{n,in}$
ADD $R_n, \#value$	T4	$R_{n,out}$, Y_{in}
	T5	$IR(immediate)_{out}$, SelectY, Add, Z_{in}
	T6	Z_{out} , $R_{n,in}$
ADD R_n, R_n	T4	$R_{n,out}$, Y_{in}
	T5	$R_{n,out}$, SelectY, Add, Z_{in}
	T6	Z_{out} , $R_{n,in}$
BZ (R_n)	T4	$R_{n,out}$, PC_{in} (if $Z = 1$)

Determine where each control signal appears

Control Signal	Instruction / Cycle	Step(s)
Y_{in}	ADD $R_n, off(R_n)$	T4, T7
	ADD $R_n, \#value$	T4
	ADD R_n, R_n	T4
Z_{in}	Fetch cycle	T1
	ADD $R_n, off(R_n)$	T5, T8
	ADD $R_n, \#value$	T5
	ADD R_n, R_n	T5
$R_{n,in}$	ADD $R_n, off(R_n)$	T9
	ADD $R_n, \#value$	T6
	ADD R_n, R_n	T6
MAR_{in}	Fetch cycle	T1
	ADD $R_n, off(R_n)$	T6

Opcode signals notation

Signal	Instruction
D_1	ADD $R_n, \text{off}(R_n)$
D_2	ADD $R_n, \# \text{value}$
D_3	ADD R_n, R_n
D_4	BZ (R_n)

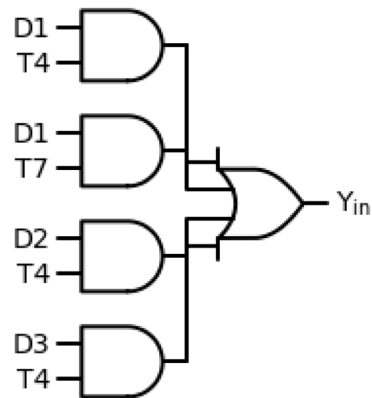
Boolean Equations for Control Signals

Control Signal	Boolean Equation
$\mathbf{Y_{in}}$	$Y_{in} = (D_1 \cdot T_4) + (D_1 \cdot T_7) + (D_2 \cdot T_4) + (D_3 \cdot T_4)$
$\mathbf{Z_{in}}$	$Z_{in} = T_1 + (D_1 \cdot T_5) + (D_1 \cdot T_8) + (D_2 \cdot T_5) + (D_3 \cdot T_5)$
$R_{n,in}$	$R_{n,in} = (D_1 \cdot T_9) + (D_2 \cdot T_6) + (D_3 \cdot T_6)$
$\mathbf{MAR_{in}}$	$MAR_{in} = T_1 + (D_1 \cdot T_6)$

Control Signals Circuit Diagram

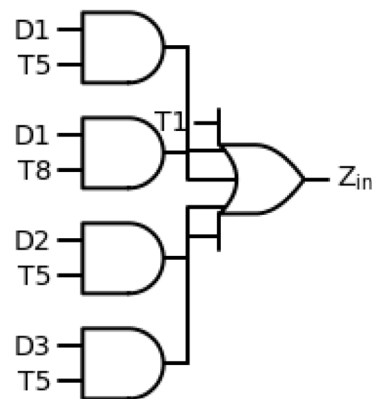
1. Generating of The Y_{in} Control Signal

$$Y_{in} = (D_1 \cdot T_4) + (D_1 \cdot T_7) + (D_2 \cdot T_4) + (D_3 \cdot T_4)$$



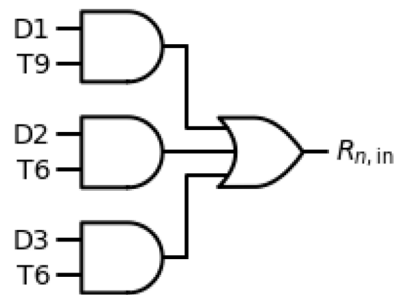
2. Generating of The Z_{in} Control Signal

$$Z_{in} = T_1 + (D_1 \cdot T_5) + (D_1 \cdot T_8) + (D_2 \cdot T_5) + (D_3 \cdot T_5)$$



3. Generating of The $R_{n, in}$ Control Signal

$$R_{n, in} = (D_1 \cdot T_9) + (D_2 \cdot T_6) + (D_3 \cdot T_6)$$



4. Generating of The MAR_{in} Control Signal

$$MAR_{in} = T_1 + (D_1 \cdot T_6)$$

